

```

import java.io.IOException;

import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;

import org.apache.hadoop.io.FloatWritable;
import org.apache.hadoop.io.Text;

import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.Mapper;
import org.apache.hadoop.mapreduce.Reducer;

import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class WeatherAnalysis {

    public static class WeatherMapper
        extends Mapper<Object, Text, Text, FloatWritable> {

        public void map(Object key, Text value, Context context)
            throws IOException, InterruptedException {

            String[] data = value.toString().split(" ");

            float temp = Float.parseFloat(data[1]);
            float dew = Float.parseFloat(data[2]);
            float wind = Float.parseFloat(data[3]);

            context.write(new Text("Temperature"), new FloatWritable(temp));
            context.write(new Text("DewPoint"), new FloatWritable(dew));
            context.write(new Text("WindSpeed"), new FloatWritable(wind));
        }
    }

    public static class WeatherReducer
        extends Reducer<Text, FloatWritable, Text, FloatWritable> {

        public void reduce(Text key, Iterable<FloatWritable> values,
            Context context)
            throws IOException, InterruptedException {

            float sum = 0;
            int count = 0;

            for (FloatWritable val : values) {
                sum += val.get();
                count++;
            }
        }
    }
}

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    }

    float avg = sum / count;
    context.write(key, new FloatWritable(avg));
}
}

public static void main(String[] args) throws Exception {

    Configuration conf = new Configuration();
    Job job = Job.getInstance(conf, "Weather Analysis");

    job.setJarByClass(WeatherAnalysis.class);
    job.setMapperClass(WeatherMapper.class);
    job.setReducerClass(WeatherReducer.class);

    job.setOutputKeyClass(Text.class);
    job.setOutputValueClass(FloatWritable.class);

    FileInputFormat.addInputPath(job, new Path(args[0]));
    FileOutputFormat.setOutputPath(job, new Path(args[1]));

    System.exit(job.waitForCompletion(true) ? 0 : 1);
}
}

```

Input:

```

2023 30 20 10
2023 32 22 12
2023 28 18 8

```

Fromat: Year Temperature DewPoint WindSpeed

```

        WRONG_LENGTH=0
        WRONG_MAP=0
        WRONG_REDUCE=0
    File Input Format Counters
        Bytes Read=42
    File Output Format Counters
        Bytes Written=58

C:\weatherproject>type output2\part-r-00000
DewPoint      20.0
Temperature    30.0
WindSpeed      10.0

```